# Mock B · Paper 1: Wrong-Answer Notes

CFA Level II · 23 wrong answers, one per page

**21 / 44**

Paper 1 (48%)

### Score by topic

| Topic | Score | % |
|---|---|---|
| Quant | 0/4 | 0% |
| FI | 1/4 | 25% |
| Corp | 2/4 | 50% |
| FSA | 2/4 | 50% |
| PM | 4/8 | 50% |
| Alt | 2/4 | 50% |
| Equity | 5/8 | 62% |
| Ethics | 5/8 | 62% |

### Questions at a glance

| Q | Topic | You | Ans |
|---|---|---|---|
| Q3 | Corp | B | C |
| Q4 | Corp | C | A |
| Q6 | FSA | A | B |
| Q8 | FSA | A | B |
| Q9 | PM | C | B |
| Q10 | PM | C | A |
| Q12 | PM | B | C |
| Q15 | Equity | A | B |
| Q16 | Equity | A | B |
| Q17 | FI | B | A |
| Q19 | FI | C | A |
| Q20 | FI | C | A |
| Q22 | Alt | C | A |
| Q24 | Alt | C | B |
| Q25 | Quant | B | C |
| Q26 | Quant | B | A |
| Q27 | Quant | B | C |
| Q28 | Quant | A | C |
| Q31 | Equity | C | B |
| Q35 | PM | A | B |
| Q38 | Ethics | C | A |
| Q40 | Ethics | C | A |
| Q44 | Ethics | B | A |

**Mock B · Paper 1 · Q3** You **B** · Answer **C**

# Corp

VIGNETTE (RELEVANT PART, UNEDITED)

Singh would like to take advantage of the under-leveraged balance sheet of Axiom while ensuring that the company's current shareholders retain control. Singh considers what corporate actions could achieve these objectives.

**Question 3. Which of the following corporate actions would best meet Singh's objectives?**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | Reorganization | | A reorganization is a court-supervised restructuring of an insolvent company. Axiom is under-leveraged and Singh wants to add debt, so this does not fit. |
| **B** | Leveraged buyout | ✗ Your pick | An LBO is an acquirer-led strategy that uses a lot of debt to buy a target. Singh wants the current shareholders to keep control, so this does not fit. |
| **C** | Dividend recapitalization | ✓ Correct | A dividend recapitalization restructures the mix from equity to debt through debt-financed dividends or repurchases. It replaces expensive equity with cheaper debt, does not dilute shareholders' stakes and can increase value to shareholders. |

**Trigger** “under-leveraged balance sheet”
“current shareholders retain control”

**Rule next time: Under-leveraged + shareholders must keep control + replace equity by debt = dividend recapitalization (debt-financed dividend or buyback). Reorganization = court-supervised insolvency; LBO = an acquirer buys the firm.**

**Key line:** Dividend recap: debt up, equity down, stakes not diluted, WACC down.

**Mock B · Paper 1 · Q4** You **C** · Answer **A**

# Corp

VIGNETTE (RELEVANT PART, UNEDITED)

Division B generated an EBITDA of $10.8 million last year and is valued at 6.0x EV/EBITDA. Division B has two segments:

A wholesale segment which contributed 20% to the division's EBITDA. Comparable wholesale retailers are valued at 7.0x EV/EBITDA.

A consumer segment which contributed 80% to the division's EBITDA. Comparable consumer retailers are valued at 5.1x EV/EBITDA.

Singh evaluates if spinning off Division B's wholesale segment would have a material impact on shareholder value, which she defines as a change in shareholder value of greater than 5%.

**Question 4. Based on Singh's analysis, spinning off the wholesale segment for Division B would lead to:**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | a material decrease in shareholder wealth. | ✓ Correct | Sum of the parts: 7.0 × 20% × 10.8 = 15.120 and 5.1 × 80% × 10.8 = 44.064, total 59.184. Division B is valued at 6 × 10.8 = 64.800, so the change is (59.184 − 64.800)/64.800 = −8.7%, a material decrease (the market values synergies between the segments). |
| **B** | no material impact on shareholder wealth. | | Incorrect: this averages the two multiples ((7.0 + 5.1)/2 = 6.05), giving 65.340 and a change of only +0.8%. |
| **C** | a material increase in shareholder wealth. | ✗ Your pick | Incorrect: this swaps the multiples (5.1 × 20% × 10.8 + 7.0 × 80% × 10.8 = 71.496), giving +10.3%. |

**Trigger** “valued at 6.0x EV/EBITDA”
“contributed 20% to the division's EBITDA”
“change in shareholder value of greater than 5%”

**Rule next time: Sum of the parts: each segment's EBITDA × ITS OWN comparable multiple, then compare with the combined EV. Material = change above ±5%.**

**Key line:** SOTP = 7.0 × 20% × 10.8 + 5.1 × 80% × 10.8 = 59.184; combined = 6.0 × 10.8 = 64.8; change = −8.7%.

**Mock B · Paper 1 · Q6** You **A** · Answer **B**

# FSA

VIGNETTE (RELEVANT PART, UNEDITED)

On 1 January of the current year, Lonza granted 4 million stock options to employees. The options vest 3 years after the grant date and expire 6 years after the grant date. The fair value per option was EUR 1.8 on the grant date.

**Question 6. For the current year, Lonza's compensation expense related to its stock options will be:**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | EUR 1.2 million. | ✗ Your pick | Incorrect: this divides the total fair value by the 6-year term to expiry instead of the vesting period. |
| **B** | EUR 2.4 million. | ✓ Correct | Expense = (4 million × EUR 1.8) / 3-year vesting period = EUR 2.4 million. |
| **C** | EUR 7.2 million. | | Incorrect: this recognizes the whole grant-date fair value in one year, without spreading it over the vesting period. |

**Trigger** “vest 3 years after the grant date”
“expire 6 years after the grant date”

**Rule next time: Option expense is spread over the VESTING period, not the term to expiry. One year's expense = grant-date fair value × options ÷ vesting years.**

**Key line:** 4m × EUR1.8 ÷ 3 = EUR2.4m.

**Mock B · Paper 1 · Q8** You **A** · Answer **B**

# FSA

VIGNETTE (RELEVANT PART, UNEDITED)

**Exhibit 2**

| | Most Recent Year | Prior Year |
|---|---|---|
| Liquidity Coverage Ratio | 95% | 110% |
| Net Stable Funding Ratio | 95% | 90% |

**Question 8. Based on Lonza's liquidity ratios at the end of the most recent year, which of the following is correct?**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | Lonza's Net Stable Funding Ratio has deteriorated from the prior year | ✗ Your pick | Incorrect: a higher NSFR is an improvement and it rose from 90% to 95%. |
| **B** | Lonza's required stable funding is higher than its available stable funding | ✓ Correct | NSFR = available stable funding / required stable funding. At 95% (below 100%), required stable funding is higher than available stable funding. |
| **C** | Lonza can withstand cash outflows 5% higher than its 30-day liquidity needs in a stress scenario | | Incorrect: an LCR of 95% means Lonza can cover only 95% of its 30-day stress outflows, i.e. 5% less, not 5% more. |

**Trigger** “Liquidity Coverage Ratio | 95% | 110%”
“Net Stable Funding Ratio | 95% | 90%”

**Rule next time: NSFR = available ÷ required stable funding: below 100% means required > available; a higher NSFR than last year = improvement. LCR below 100% = cannot cover the 30-day stress outflows (it is short, not 5% to spare).**

**Key line:** NSFR 95% < 100% → required stable funding is higher than available.

**Mock B · Paper 1 · Q9** You **C** · Answer **B**

# PM

VIGNETTE (RELEVANT PART, UNEDITED)

Grap backtests the strategy using a model with the following parameters:

• Rebalancing frequency: monthly (simulated using rolling windows);

• Formation of factor portfolios: for each factor, buy the top 20% of stocks and sell the bottom 20% of stocks ranked by the factor over the previous 12 months;

• Weighting of factor portfolios: risk parity.

**Question 9. Which of the following will Grap's backtesting model require?**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | Non-overlapping rolling windows | | Incorrect: rolling windows overlap (each has a 12-month in-sample period, 11 months of which overlap with the previous window). |
| **B** | Two iterations of the rolling-window procedure | ✓ Correct | One rolling-window iteration builds the factor portfolios and a second one over the same time span estimates the covariance matrix for risk parity weights, which avoids look-ahead bias. |
| **C** | A 13-month in-sample period for each rolling window | ✗ Your pick | Incorrect: the in-sample period is the previous 12 months (not 13) with a 1-month out-of-sample period. |

**Trigger** "Rebalancing frequency: monthly (simulated using rolling windows)"
"Weighting of factor portfolios: risk parity"

**Rule next time: Two layers (factor portfolios, then risk-parity weights from a covariance matrix) = TWO rolling-window iterations; windows overlap; the in-sample period is the previous 12 months (not 13).**

**Key line:** Layer 1: factor ranks over 12 months. Layer 2: 10-year covariance matrix → equal-risk-contribution weights.

**Mock B · Paper 1 · Q10** You **C** · Answer **A**

# PM

VIGNETTE (RELEVANT PART, UNEDITED)

The TOP100 Index is updated regularly and, as a result, its composition has significantly changed over the 10-year backtesting period. Grap wants to ensure that survivorship bias is mitigated and compares three alternative methods to form factor portfolios in each rolling window:

- Method A only uses the companies that were in the TOP100 Index at any given point in time t.
- Method B only uses the companies that were in the TOP100 Index at the end of the 10-year period.
- Method C only uses the companies that have been in the TOP100 Index for the entire 10-year period.

**Question 10. To mitigate survivorship bias, which method should Grap use when forming factor portfolios?**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | Method A | ✓ Correct | Point-in-time data: in each rolling window use only the companies that were in the index at that time (casualties as well as survivors). |
| **B** | Method B | | Incorrect: only companies in the index at the end of the period are survivors, which creates survivorship bias. |
| **C** | Method C | ✗ Your pick | Incorrect: only companies in the index for the entire 10 years are survivors, which creates survivorship bias. |

**Trigger** “composition has significantly changed”
“Method A only uses the companies that were in the TOP100 Index at any given point in time t.”

**Rule next time: Survivorship bias is avoided with point-in-time data (members as they were at each date, including later casualties). Methods that start from today's members, or from names present throughout, keep only survivors.**

**Key line:** Point-in-time universe = no survivorship bias.

**Mock B · Paper 1 · Q12** You **B** · Answer **C**

# PM

VIGNETTE (RELEVANT PART, UNEDITED)

Grap performs a complementary risk analysis using a risk measure that can give him a better understanding of the downside risk of the absolute return trading strategy to be employed by the equity market-neutral fund. The strategy is benchmarked against cash.

**Question 12. For his complementary risk analysis, which of the following risk measures should Grap use?**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | Relative VaR | | Incorrect: with a cash benchmark (zero volatility) relative VaR equals VaR, so it adds no information. |
| **B** | Surplus at risk | ✗ Your pick | Incorrect: surplus at risk is for pension funds with liabilities, not for a hedge fund. |
| **C** | Maximum drawdown | ✓ Correct | For an absolute return, equity market-neutral strategy, maximum drawdown (worst peak-to-trough decline) complements standard deviation and beta as a downside risk measure. |

**Trigger** “absolute return trading strategy”
“benchmarked against cash”
“better understanding of the downside risk”

**Rule next time: Absolute return vs a cash benchmark: relative VaR equals plain VaR (adds nothing); surplus at risk is for pension liabilities; downside of a market-neutral strategy = maximum drawdown (worst peak-to-trough).**

**Key line:** Cash benchmark has zero volatility → relative VaR = VaR.

**Mock B · Paper 1 · Q15** You **A** · Answer **B**

# Equity

VIGNETTE (RELEVANT PART, UNEDITED)

Kassam evaluates the stock of Altair, which is a recently listed, high growth, biotechnology company. She is considering a stable market-based multiple to value the stock and does not want to make any adjustments to the reported financial statements. After reviewing recent analyst reports, Kassam notices that Altair has internally generated brands that are not shown as intangible assets on Altair's balance sheet. Further, she notes that certain discretionary accounting decisions about expenses have led to an inaccurate representation of Altair's economic performance.

**Question 15. Which of the following valuation ratios is most suitable for Kassam to value Altair?**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | Price-to-book ratio | ✗ Your pick | Incorrect: P/B misses internally generated intangibles such as brands, which are not on the balance sheet. |
| **B** | Price-to-sales ratio | ✓ Correct | Sales are less subject to distortion or manipulation than EPS or book value and are more stable than EPS, so P/S is suitable without adjusting reported financials. |
| **C** | Price-to-earnings ratio | | Incorrect: earnings are prone to manipulation through discretionary expense decisions and distort EPS. |

**Trigger** “stable market-based multiple”
“does not want to make any adjustments to the reported financial statements”
“internally generated brands”

**Rule next time: No adjustments wanted + unreliable earnings and book value (internal brands, discretionary expenses) → P/S: sales are hard to manipulate and more stable than EPS. P/B misses internal intangibles; P/E is distorted by expense choices.**

**Key line:** P/S: stable, less manipulable, no adjustments needed.

**Mock B · Paper 1 · Q16** You **A** · Answer **B**

# Equity

VIGNETTE (RELEVANT PART, UNEDITED)

Kassam values Becq, a small private company, using the excess earnings method. Relevant financial information is shown in Exhibit 3.

**Exhibit 3**

| | |
|---|---|
| Fixed assets | $725,000 |
| Working capital | $315,000 |
| Last reported earnings | $191,290 |
| Residual income growth rate | 2.50% |
| Required return on fixed assets | 8.50% |
| Required return on working capital | 3.25% |
| Discount rate | 11.25% |

**Question 16. Based on Exhibit 3, the value of Becq using the excess earnings method is closest to:**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | $1,400,000. | ✗ Your pick | Incorrect: this is only the value of the intangibles (1,399,008); tangible assets must be added. |
| **B** | $2,440,000. | ✓ Correct | Residual earnings = 191,290 − 315,000 × 3.25% − 725,000 × 8.50% = 119,427.50. Intangibles = 119,427.50 × 1.025/(11.25% − 2.50%) = 1,399,008. Total = 1,399,008 + 315,000 + 725,000 = about 2,440,000. |
| **C** | $3,080,000. | | Incorrect: the required return on fixed assets (8.50%) is used as the capitalization rate instead of the discount rate. |

**Trigger** “excess earnings method”
“Required return on working capital | 3.25%”
“Discount rate | 11.25%”

**Rule next time: Excess earnings: residual = earnings − required returns on working capital and fixed assets; capitalise it with the DISCOUNT rate − g (not the asset return) to value the intangibles, then ADD the tangible assets for the whole business.**

**Key line:** 119,427.5 × 1.025 ÷ (11.25% − 2.5%) = 1,399,008; + 315,000 + 725,000 = 2.44m.

**Mock B · Paper 1 · Q17** You **B** · Answer **A**

# FI

VIGNETTE (RELEVANT PART, UNEDITED)

She first considers the following 2-year, default-free bonds with embedded options:

Bond A is a 5% annual pay coupon bond and is callable at par ($100) at the end of year 1.

To analyze these bonds, Plessinger creates a binomial interest rate tree using the rates in Exhibit 1, assuming an interest rate volatility of 20%, with branching from each node to a high state and a low state occurring with equal probability. Plessinger also considers the effective convexity of Bond B.

**Exhibit 1 Binomial Interest Rate Tree (1-year rates)**

| Year 0 | Year 1 | Year 2 | |
|---|---|---|---|
| Upper node | | 6.09% | 7.32% |
| Middle node | 2.00% | | 4.91% |
| Lower node | | 4.08% | 3.29% |

Note: This exhibit was a binomial diagram in the original paper. Its figure was not in the pasted text: the Year 0 and Year 1 rates come from the answer explanations, the Year 2 nodes are reconstructed from them (4.08% × e^0.4 = 6.09%, 3.29% × e^0.4 = 4.91% and 3.29% × e^0.8 = 7.32%).

**Question 17. Based on the binomial interest rate tree, the value of Bond A is closest to:**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | $102.44. | ✓ Correct | Work backwards: at 6.09% the value is 105/1.0609 = 98.97; at 4.08% it is 105/1.0408 = 100.88, which exceeds par so the bond is called at 100. Year 0: (100 + 5)/1.02 = 102.9412 and (98.97 + 5)/1.02 = 101.9339; the average is 102.44. |
| **B** | $102.87. | ✗ Your pick | Incorrect: this ignores the call (uses 100.88 instead of 100): 102.87. |
| **C** | $103.37. | | Incorrect: this treats the bond as putable instead of callable (103.37). |

**Trigger** “Bond A is a 5% annual pay coupon bond and is callable at par”
“interest rate volatility of 20%”

**Rule next time: Callable bond on the tree: work backwards from the end; at any node where the value exceeds the call price (100) the issuer calls, so cap the value at 100; then average the two nodes and discount with the node rate.**

**Key line:** Year 1: 105/1.0609 = 98.97 and min(105/1.0408, 100) = 100; (100+5)/1.02 and (98.97+5)/1.02 → 102.44.

**Mock B · Paper 1 · Q19** You **C** · Answer **A**

# FI

VIGNETTE (RELEVANT PART, UNEDITED)

Plessinger next evaluates a 3-year maturity, callable investment-grade corporate bond. She computes an OAS of 51 bps using the same binomial interest rate tree in Exhibit 1. Plessinger considers the impact on the bond's OAS under a scenario where the bond's price is unchanged and interest rate volatility increases from 20% to 25%.

**Question 19. Under Plessinger's scenario, the OAS of the corporate bond will:**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | decrease. | ✓ Correct | Higher volatility raises the value of the embedded call. With the price (and Z-spread) fixed, and Z-spread = OAS + option cost, the OAS must decline. |
| **B** | not change. | | Incorrect: higher volatility decreases the OAS of a callable bond. |
| **C** | increase. | ✗ Your pick | Incorrect: higher volatility decreases the OAS of a callable bond. |

**Trigger** “callable investment-grade corporate bond”
“interest rate volatility increases from 20% to 25%”
“the bond's price is unchanged”

**Rule next time: Callable bond, price fixed, volatility up: the call option is worth more; Z-spread = OAS + option cost, so the OAS FALLS (for a putable bond it would rise).**

**Key line:** Z-spread = OAS + option cost.

**Mock B · Paper 1 · Q20** You **C** · Answer **A**

# FI

VIGNETTE (RELEVANT PART, UNEDITED)

**Exhibit 1 Binomial Interest Rate Tree (1-year rates)**

| Year 0 | Year 1 | Year 2 | |
|---|---|---|---|
| Upper node | | 6.09% | 7.32% |
| Middle node | 2.00% | | 4.91% |
| Lower node | | 4.08% | 3.29% |

Note: This exhibit was a binomial diagram in the original paper. Its figure was not in the pasted text: the Year 0 and Year 1 rates come from the answer explanations, the Year 2 nodes are reconstructed from them (4.08% × e^0.4 = 6.09%, 3.29% × e^0.4 = 4.91% and 3.29% × e^0.8 = 7.32%).

Finally, still using the binomial interest rate tree in Exhibit 1, Plessinger analyzes a 3-year, default-free capped floating rate bond with a par value of $100. The bond pays an annual coupon equal to the 1-year reference rate, is set in arrears, and is capped at 5.0%.

**Question 20. The value of the capped floating rate bond analyzed by Plessinger is:**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | less than its par value. | ✓ Correct | The cap of 5% binds when rates are above 5% (the 6.09% Year 1 rate and the 7.32% Year 2 rate), so there are paths where the floater's coupon is limited and its value is below par. |
| **B** | equal to its par value. | | Incorrect: without the cap a floater sells at par on each reset date, but the cap binds on some paths in the tree even though the spot rates are below 5%. |
| **C** | greater than its par value. | ✗ Your pick | Incorrect: a cap can only reduce the value of a floater. |

**Trigger** “capped floating rate bond”
“capped at 5.0%”
“set in arrears”

**Rule next time: A cap limits the coupon on paths where the tree rate exceeds the cap (6.09% in Year 1, 7.32% in Year 2), so the capped floater is worth LESS than par; without a cap a floater trades at par on reset dates.**

**Key line:** Cap binds on some nodes → value < par.

**Mock B · Paper 1 · Q22** You **C** · Answer **A**

# Alt

VIGNETTE (RELEVANT PART, UNEDITED)

Chang is also evaluating hedge funds as potential additions to the portfolio. She identifies the following strategies and considers their returns, leverage, and liquidity:

**Strategy 1: Multi-strategy funds**

**Strategy 2: Fund of funds**

**Question 22. Compared to Strategy 2, Strategy 1 most likely has:**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | higher returns. | ✓ Correct | Multi-strategy funds have generally outperformed funds of funds, but with more variance and occasional large losses linked to higher leverage; flexibility and fast tactical allocation help. |
| **B** | lower leverage. | | Incorrect: multi-strategy funds use significantly more leverage than most funds of funds. |
| **C** | fewer gates on redemptions. | ✗ Your pick | Incorrect: multi-strategy funds often impose investor-level or fund-level gates, unlike funds of funds. |

**Trigger** “Strategy 1: Multi-strategy funds”
“Strategy 2: Fund of funds”
“returns, leverage, and liquidity”

**Rule next time: Multi-strategy vs fund of funds: multi-strategy = higher returns, MORE leverage, more variance and more gates; funds of funds = more diversified mix, lower leverage, somewhat lower variance.**

**Key line:** Multi-strategy: return up, leverage up, gates yes.

**Mock B · Paper 1 · Q24** You **C** · Answer **B**

# Alt

VIGNETTE (RELEVANT PART, UNEDITED)

The pension plan has an objective to limit drawdowns. Chang considers what individual hedge fund strategy would best meet that objective when combined with the pension plan's stock and bond portfolio.

**Question 24. Which of the following hedge fund strategies is best suited to meet the plan's drawdown objective?**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | Event-driven | | Incorrect: event-driven/distressed strategies show high maximum drawdowns when combined with a stock and bond portfolio and little risk reduction. |
| **B** | Global macro | ✓ Correct | The opportunistic strategies (global macro, systematic futures, merger arbitrage, equity market-neutral) give the smallest maximum drawdowns when combined with a traditional portfolio and have low serial autocorrelation (good liquidity). |
| **C** | Relative value | ✗ Your pick | Incorrect: relative value/convertible arbitrage strategies show high maximum drawdowns and high standard deviations when combined. |

**Trigger** “limit drawdowns”
“combined with the pension plan's stock and bond portfolio”

**Rule next time: Smallest maximum drawdown when added to a stock and bond portfolio: opportunistic strategies (global macro, systematic futures, merger arbitrage, equity market-neutral). Event-driven, relative value and long/short equity show high drawdowns.**

**Key line:** Low drawdown contribution → global macro.

**Mock B · Paper 1 · Q25** You **B** · Answer **C**

# Quant

VIGNETTE (RELEVANT PART, UNEDITED)

Moreno is tasked with a new project that looks to classify bond issuers based on their probability of default within the next 12 months. He gathers a large sample of financial and accounting metrics for 1,000 issuers and observes that the issuers are not linearly separable. He compares non-linear support vector machine (SVM), unconstrained classification and regression tree (CART), and SVM with soft margin classification. He wants to use a machine-learning algorithm that addresses non-linear separability and is not prone to overfitting.

**Question 25. Which of the following machine-learning algorithms is most suitable for Moreno's bond issuer classification model?**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | Non-linear SVM | | Incorrect: non-linear SVM with more advanced boundaries reduces training misclassifications but is complex and prone to overfitting. |
| **B** | Unconstrained CART | ✗ Your pick | Incorrect: unconstrained CART can perfectly learn the training data and overfit unless regularization parameters are added. |
| **C** | SVM with soft margin classification | ✓ Correct | SVM with soft margin classification adds a penalty for misclassified training observations and trades off a wider margin against total error, handling non-linear separability without overfitting. |

**Trigger** “not linearly separable”
“addresses non-linear separability and is not prone to overfitting”

**Rule next time: Non-linear separability + no overfitting → SVM with soft margin classification (penalty for misclassified points, wider margin). Non-linear SVM and unconstrained CART overfit.**

**Key line:** Soft margin: trade-off wider margin vs total error penalty.

**Mock B · Paper 1 · Q26** You **B** · Answer **A**

# Quant

VIGNETTE (RELEVANT PART, UNEDITED)

Moreno also creates Model 1, a logistic regression model that uses two factors, profit margin (PROFIT) and leverage ratio (LEVER), to predict the probability of default. The binary dependent variable is 1 for companies that defaulted within the subsequent 12 months and 0 otherwise. Using the regression results in Exhibit 1, Moreno calculates the predicted probability of default for the average distressed company, which has PROFIT = –2.0 and LEVER = 3.0.

**Exhibit 1**

| | Coefficient |
|---|---|
| Intercept | –1.7878 |
| PROFIT | –0.0185 |
| LEVER | 0.3876 |

**Question 26. For the average distressed company, the probability of default predicted by Model 1 is closest to:**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | 36%. | ✓ Correct | P = 1/[1 + exp(−(−1.7878 − 0.0185 × (−2.0) + 0.3876 × 3.0))] = 1/(1 + 1.8004) = 0.357, about 36%. |
| **B** | 56%. | ✗ Your pick | Incorrect: 0.56 is exp(b0 + b1PROFIT + b2LEVER) = exp(−0.5880), the odds, not the probability. |
| **C** | 77%. | | Incorrect: 77% omits the intercept. |

**Trigger** “PROFIT = –2.0 and LEVER = 3.0”
“Intercept | –1.7878”

**Rule next time: Logit: P = 1 ÷ (1 + exp(−(b0 + b1x1 + b2x2))). exp(linear part) alone is the ODDS, not the probability, and dropping the intercept gives 77%.**

**Key line:** x = −1.7878 + 0.0370 + 1.1628 = −0.588; P = 1 ÷ (1 + e^0.588) = 0.357.

**Mock B · Paper 1 · Q27** You **B** · Answer **C**

# Quant

VIGNETTE (RELEVANT PART, UNEDITED)

To analyze the drivers of ROE for software companies, Moreno collects fundamental data and compares two linear regression models. Model 2 uses profit margin (PROFIT) and interest coverage (COVER). Model 3 extends Model 2 by adding a third factor, the quick ratio (QUICK).

**Exhibit 2**

| Coefficient | t-Statistic | p-Value | |
|---|---|---|---|
| Intercept | –1.4411 | –3.4888 | 0.0000 |
| PROFIT | 0.4840 | 49.5319 | 0.0000 |
| COVER | 0.7487 | 27.9215 | 0.0000 |
| QUICK | 0.0919 | 1.9204 | 0.0558 |

**Question 27. Compared to Model 2, Model 3 most likely has a:**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | lower R-squared and a higher adjusted R-squared. | | Incorrect: $R^2$ never decreases when a variable is added. |
| **B** | higher R-squared and a lower adjusted R-squared. | ✗ Your pick | Incorrect: adjusted $R^2$ rises when the added coefficient's $\|t\| > 1.0$, and QUICK's t-statistic is 1.9204. |
| **C** | higher R-squared and a higher adjusted R-squared. | ✓ Correct | $R^2$ increases or stays the same when a variable is added, and adjusted $R^2$ increases if the new coefficient's \|t-statistic\| is above 1.0 (QUICK: 1.9204). |

**Trigger** “adding a third factor, the quick ratio (QUICK)”
“QUICK | 0.0919 | 1.9204 | 0.0558”

**Rule next time: Adding a variable never lowers $R^2$; adjusted $R^2$ rises if the new coefficient's $|t| > 1.0$ (QUICK: 1.92), even though p = 0.0558 is not significant at 5%.**

**Key line:** $|t| > 1 \rightarrow$ adjusted $R^2$ up; $|t| < 1 \rightarrow$ down.

**Mock B · Paper 1 · Q28** You **A** · Answer **C**

# Quant

VIGNETTE (RELEVANT PART, UNEDITED)

Finally, Moreno wants to test if last-12-month sales (SALES) are a good predictor of ROE. He runs Model 4, which has the following specification:

A plot of the regression residuals against SALES is presented in Exhibit 3.

Note: Exhibit 3 was an image in the original paper (file KL3.png) and was not in the pasted text. From the answer explanation: in the plot of the Model 4 residuals against SALES, the variance of the residuals increases as SALES increase.

**Question 28. The most appropriate conclusion to draw from the residual plot for Model 4 is that:**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | the model suffers from unconditional heteroskedasticity. | ✗ Your pick | Incorrect: the residual variance increases with SALES, so it is correlated with an independent variable: conditional, not unconditional, heteroskedasticity. |
| **B** | there is a non-linear relationship between ROE and SALES. | | Incorrect: a residual plot does not show the form of the relationship between target and feature. |
| **C** | the F-test for the overall regression significance is unreliable. | ✓ Correct | With conditional heteroskedasticity the MSE is a biased estimator, so the F-test for overall significance is unreliable (and Type I errors rise). |

**Trigger** “A plot of the regression residuals against SALES”
“the variance of the residuals increases as SALES increase”
**Rule next time: Residual variance that grows with the regressor = CONDITIONAL heteroskedasticity: biased standard errors, unreliable F-test, more Type I errors. Unconditional heteroskedasticity is not a problem. A residual plot does not show non-linearity of the relationship.**

**Key line:** Fan-shaped residuals → conditional heteroskedasticity.

**Mock B · Paper 1 · Q31** You **C** · Answer **B**

# Equity

VIGNETTE (RELEVANT PART, UNEDITED)

McCormick reviews his FCFE valuation model for Firm C. Firm C's management has just announced that they will issue new common stock during the next year. McCormick evaluates the potential effect of this change on next year's FCFE forecast.

**Question 31. Given Firm C's announced change, next year's FCFE forecast will most likely:**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | decrease. | | Incorrect: transactions between the company and its shareholders (dividends, repurchases, issuances) do not affect free cash flow. |
| **B** | remain unchanged. | ✓ Correct | Share issuances are a use of FCFE, not a determinant of it, so the forecast is unchanged. |
| **C** | increase. | ✗ Your pick | Incorrect: share issuance does not affect free cash flow. |

**Trigger** “will issue new common stock during the next year”
“next year's FCFE forecast”

**Rule next time: Share issuance, repurchases and dividends are USES of FCFE, not drivers of it: next year's FCFE is unchanged. FCFE changes only with CFO, capex and net borrowing.**

**Key line:** FCFE = CFO − FCInv + net borrowing.

**Mock B · Paper 1 · Q35** You **A** · Answer **B**

# PM

VIGNETTE (RELEVANT PART, UNEDITED)

Chu reviews a diversified US bond portfolio, Portfolio A. His objective is to determine which of the following three sectors' spreads can be expected to narrow the most if the economy recovers from a business downturn.

**Sector A: Investment grade consumer cyclical**

**Sector B: Speculative grade consumer cyclical**

**Sector C: Speculative grade consumer noncyclical**

**Question 35. For Portfolio A, which of the following sectors most likely satisfies Chu's objective concerning credit spreads?**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | Sector A | ✗ Your pick | Incorrect: investment grade cyclical spreads narrow less than speculative grade cyclical spreads. |
| **B** | Sector B | ✓ Correct | Sector B is both speculative grade and cyclical, so it is the most sensitive to an improving economy and its spreads narrow the most. |
| **C** | Sector C | | Incorrect: noncyclical issuers are less sensitive to the business cycle than cyclical issuers. |

**Trigger** “spreads can be expected to narrow the most”
“economy recovers from a business downturn”
**Rule next time: Spreads move most for low-rated AND cyclical issuers: they widen most in a downturn and narrow most in a recovery → speculative grade + cyclical = Sector B.**

**Key line:** Low rating + cyclical = biggest spread swings.

**Mock B · Paper 1 · Q38** You **C** · Answer **A**

# Ethics

VIGNETTE (RELEVANT PART, UNEDITED)

Silva conducts due diligence on a mid-cap packaging company whose stock is actively traded. He determines the company's fundamentals are strong and buys its stock for one of the equity funds he manages. Silva provides commentary on the company's prospects in his monthly client newsletter. Silva has owned the company's stock in his personal portfolio for many years and has disclosed this in the fund's communications to clients. A few weeks after releasing the newsletter, Silva sells some of his personal position in the stock as part of rebalancing his personal portfolio. He earns a substantial profit from this sale.

**Question 38. Did Silva violate the Standards when selling his personal stake in the packaging company?**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | No | ✓ Correct | Standard VI(B) Priority of Transactions: client transactions have priority. The fund bought first and selling an actively traded stock from his personal portfolio did not disadvantage clients. |
| **B** | Yes, because he sold the shares at a profit | | Incorrect: a profit is not a violation as long as clients are not disadvantaged. |
| **C** | Yes, by acting against his own recommendation | ✗ Your pick | Incorrect: the Standard does not prohibit personal transactions that differ from current recommendations as long as clients are not disadvantaged. |

**Trigger** “buys its stock for one of the equity funds he manages”
“Silva sells some of his personal position”
“as part of rebalancing his personal portfolio”

**Rule next time: VI(B) Priority of Transactions: client trades come first; personal trades are allowed if clients are not disadvantaged (the fund bought first, the stock is actively traded). A profit, or trading against his own recommendation, is not a violation by itself.**

**Key line:** No client disadvantage → no violation.

**Mock B · Paper 1 · Q40** You **C** · Answer **A**

# Ethics

VIGNETTE (RELEVANT PART, UNEDITED)

Tritium's recommended policy for record retention is to maintain records of investment actions and client communications in electronic format. Silva is accustomed to retaining all of his and his clients' relevant files in hard copy form. He does not store any of the records in electronic format. He saves his records for 5 years, after which they are destroyed. Local regulations require record retention for 5 years.

**Question 40. Does Silva violate the Standards with respect to record retention?**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | No | ✓ Correct | Standard V(C): records may be kept in hard copy or electronic form, and following the regulatory requirement (5 years) satisfies the Standard; 7 years applies only absent regulation or firm policy. |
| **B** | Yes, because of the format in which the records are kept | | Incorrect: the Standard does not require a specific format. |
| **C** | Yes, because records must be retained for a minimum period of 7 years | ✗ Your pick | Incorrect: the 7-year recommendation applies only if there is no regulatory requirement; the local rule is 5 years. |

**Trigger** “hard copy form”
“He saves his records for 5 years”
“Local regulations require record retention for 5 years”

**Rule next time: V(C): records may be kept in hard copy or electronic form, for the period required by regulation or firm policy; the 7-year recommendation applies only when neither exists.**

**Key line:** Format free; period = regulation (5 years) ≥ OK.

**Mock B · Paper 1 · Q44** You **B** · Answer **A**

# Ethics

VIGNETTE (RELEVANT PART, UNEDITED)

Murphy's supervisor manages a large research department and, as a result, is not able to personally evaluate the conduct of all the employees in the department. The supervisor hires a deputy manager to oversee several employees and ensures the deputy manager is properly trained. The supervisor establishes the following policies to assist with compliance:

Policy 1: A monetary incentive will be offered as a reward for ethical behavior.

Policy 2: The deputy manager will be responsible for evaluating the ethical conduct of department members.

**Question 44. Does Murphy's supervisor violate the Standards in his oversight of the department?**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | No | ✓ Correct | Standard IV(C): incentives (monetary or not) that reward ethical behavior are allowed and delegating supervisory duties in a large department is reasonable. |
| **B** | Yes, by establishing Policy 1 | ✗ Your pick | Incorrect: monetary incentives for ethical behavior do not violate the Standard. |
| **C** | Yes, by establishing Policy 2 | | Incorrect: delegating supervision to a properly trained deputy is not a violation. |

**Trigger** “A monetary incentive will be offered as a reward for ethical behavior.”
“The deputy manager will be responsible for evaluating the ethical conduct of department members.”

**Rule next time: IV(C): a supervisor of a large department may delegate to trained deputies and may reward ethical behavior, even with money; responsibility remains with the supervisor.**

**Key line:** Incentive for ethics + trained delegate = compliant.